1. Show that the sliced score matching (SSM) loss can also be written as

$$L_{SSM} = \mathbb{E}_{x\sim p(x)}\mathbb{E}_{v\sim p(v)}\left[\|v^T S(x;\theta)\|^2 + 2v^T\nabla_x(v^T S(x;\theta))\right].$$

2. Briefly explain SDE.

1

$$L_{\text{SSM}} = \mathbb{E} * x \sim p(x)\mathbb{E} * v \sim p(v)\left[|v^T S(x;\theta)|^2 + 2v^T\nabla_x(v^T S(x;\theta))\right].$$

Derivation

Sliced Score Matching starts from the objective

$$L_{\text{SSM}} = \mathbb{E} * x \sim p(x)\mathbb{E} * v \sim p(v)\left[(v^T S_\theta(x))^2 * 2, \nabla_x \cdot (v, v^T S_\theta(x))\right].$$

Since (v) is a constant with respect to (x), we use the product rule:

$$\nabla_x \cdot (v, v^T S_\theta(x)) = v^T\nabla_x(v^T S_\theta(x)).$$

To see this, note that:

- (v) is constant, so $(\nabla_x(v) = 0)$.
- The divergence of a constant vector times a scalar function $(g(x))$ is simply:

$$\nabla_x \cdot (v, g(x)) = v^T\nabla_x g(x).$$

Apply this with $(g(x) = v^T S_\theta(x))$:

$$\nabla_x \cdot (v, v^T S_\theta(x)) = v^T\nabla_x(v^T S_\theta(x)).$$

Substitute back:

$$L_{\text{SSM}} = \mathbb{E} * x, v\left[(v^T S * \theta(x))^2 + 2v^T\nabla_x(v^T S_\theta(x))\right].$$

Finally,

$$(v^T S_\theta(x))^2 = |v^T S(x;\theta)|^2,$$

giving the desired form.

2

A **Stochastic Differential Equation (SDE)** describes the evolution of a random process over time. It has the general form:

$$dx = f(x, t), dt + g(x, t), dW_t,$$

where:

- ($f(x, t)$) is the **drift** term
- ($g(x, t)$) is the **diffusion** term
- (W_t) is a **Wiener process**

In diffusion models:

- The **forward SDE** gradually adds noise to data.
- The **reverse-time SDE** removes noise and generates samples.

SDEs are fundamental in score-based generative modeling because solving them backward requires learning the **score function** $(\nabla_x \log p_t(x))$.